

outputs we are assuming that the number of inputs on the left-hand side matches the number of inputs on the right-hand side and similarly for the outputs (as otherwise the identity would not make sense). The number of inputs or outputs can also be zero, so we also get for instance:

$$\bigcirc - \square = \bullet - \quad (17)$$

Recalling the definition of $|+\rangle$ and $|0\rangle$ in ZX (Eqs. (5) and (11)) we see that this simply states that $H|+\rangle = |0\rangle$.

Using the self-inverse rule Eq. (14) we can also rephrase the ‘Hadamard conjugation’ rule of Eq. (16) as a commutation rule that tells us how to ‘push’ a Hadamard gate through a dot:

$$\begin{array}{c} \square \\ \vdots \\ \square \end{array} \begin{array}{c} \curvearrowright \\ \alpha \\ \curvearrowleft \end{array} \begin{array}{c} \square \\ \vdots \\ \square \end{array} \stackrel{(14)}{=} \begin{array}{c} \square \\ \vdots \\ \square \end{array} \begin{array}{c} \curvearrowright \\ \alpha \\ \curvearrowleft \end{array} \begin{array}{c} \square \\ \vdots \\ \square \end{array} \stackrel{(16)}{=} \begin{array}{c} \square \\ \vdots \\ \square \end{array} \begin{array}{c} \curvearrowright \\ \bullet \\ \curvearrowleft \end{array} \begin{array}{c} \square \\ \vdots \\ \square \end{array} \quad (18)$$

In a slogan: *a Hadamard gate is commuted through a dot by changing its colour.*

As an application of the rules concerning Hadamards we can find a compact representation of a CZ gate in the ZX-calculus. Recall that a CZ gate is equal to a CNOT gate conjugated by Hadamards on the target qubit:

$$\begin{array}{c} \bullet \\ | \\ \bullet \end{array} = \begin{array}{c} \bullet \\ | \\ \square \oplus \square \end{array} \begin{array}{c} \square \\ \oplus \\ \square \end{array} \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \quad (19)$$

Translating the right-hand side to a ZX-diagram and simplifying using the Hadamard colour-change rule and the self-inverse rule we find:

$$\begin{array}{c} \square \\ | \\ \bullet \end{array} \stackrel{(18)}{=} \begin{array}{c} \square \\ | \\ \square \end{array} \begin{array}{c} \square \\ \oplus \\ \square \end{array} \begin{array}{c} \square \\ | \\ \square \end{array} \stackrel{(14)}{=} \begin{array}{c} \square \\ | \\ \square \end{array} \begin{array}{c} \square \\ \oplus \\ \square \end{array} \begin{array}{c} \square \\ | \\ \square \end{array} \quad (20)$$

What is interesting about this is that the rightmost diagram is symmetric in the two qubits, as the CZ gate indeed is, and that we did not need to introduce any new generators or concepts to the ZX-calculus to represent it. In fact, we already have all ingredients to represent any linear map between qubits: ZX-diagrams are *universal* (see Section 3.7).

3 ZX-diagrams

In this section we will introduce the building blocks of ZX-diagrams and discuss some of their main properties. We start by defining the basic generators of ZX-diagrams, the *spiders*, in Section 3.1. Then in Section 3.2 we will see how to compose these. We discuss the symmetries of ZX-diagrams in Section 3.3 which will allow us to treat ZX-diagrams as undirected graphs. Then Sections 3.4–3.7 discuss various topics culminating in a proof that ZX-diagrams are in fact universal and hence can represent any linear map between qubits.

3.1 Spiders

The first type of generator is what we referred to as a ‘white dot’ in the previous section. The more common name for this is the *Z-spider*. A Z-spider can have any number of inputs (wires coming in from the left) and outputs (wires coming out of the right) and has the following interpretation as a linear map:

$$\begin{array}{c} \curvearrowright \\ \alpha \\ \curvearrowleft \end{array} := |0 \cdots 0\rangle \langle 0 \cdots 0| + e^{i\alpha} |1 \cdots 1\rangle \langle 1 \cdots 1|, \quad (21)$$

Why it is called a ‘spider’ should be clear from the way the diagram looks. This is the Z-spider because it is defined with respect to the eigenbasis of the Z matrix, $|0\rangle$ and $|1\rangle$.

Similarly, what we referred to as a ‘grey dot’, is more commonly called an *X-spider*. This is defined as above, but with the Z eigenbasis replaced by the X eigenbasis $|+\rangle$ and $|-\rangle$:

$$\text{Diagram with a central circle labeled } \alpha \text{ and four external lines} := |+\cdots+\rangle\langle+\cdots+| + e^{i\alpha} |-\cdots-\rangle\langle-\cdots-|. \quad (22)$$

As it is instructive, let us explicitly write out some of the matrices that these spiders are equal to. Recall that:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

For the 1-input, 1-output Z -spider with a phase α we hence have:

$$\text{---}(\alpha)\text{---} = |0\rangle\langle 0| + e^{i\alpha} |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & e^{i\alpha} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \quad (23)$$

This matrix is the $R_Z(\alpha)$ phase gate that rotates the Bloch sphere by an angle α over the Z axis. In particular, for $\alpha = \pi$ we get the Z Pauli matrix, which indeed matches with our translation table (1). We can do a similar calculation for the 1-input, 1-output X -spider with a phase α :

$$\text{---}\textcircled{\alpha}\text{---} = |+\rangle\langle +| + e^{i\alpha} |-\rangle\langle -| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} e^{i\alpha} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1+e^{i\alpha} & 1-e^{i\alpha} \\ 1-e^{i\alpha} & 1+e^{i\alpha} \end{pmatrix} \quad (24)$$

This is the familiar $R_X(\alpha)$ phase gate, and for $\alpha = \pi$ we get the Pauli X matrix.

It is customary to drop the label inside the spider when the phase α is 0:

$$\begin{aligned} \text{Diagram 1} &= |0 \cdots 0\rangle \langle 0 \cdots 0| + |1 \cdots 1\rangle \langle 1 \cdots 1| \\ \text{Diagram 2} &= |+\cdots+\rangle \langle +\cdots+| + |-\cdots-\rangle \langle -\cdots-| \end{aligned}$$

In particular, we have:

$$\text{---}\bigcirc\text{---} = \text{---}\bullet\text{---} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (25)$$

In a ZX-diagram we present an identity matrix like the one above by a empty piece of wire, and hence we have the following equation.

$$\text{---} \bigcirc \text{---} = \text{---} = \text{---} \bullet \text{---}$$

Indeed, it is this equation we used in the proof (8).

In the previous section we claimed we could represent the Pauli Z and X basis states using spiders. We can now verify that this is indeed the case:

$$\text{---}\bullet = |+\rangle + |-\rangle = \sqrt{2}|0\rangle \quad \text{---}\circ = |0\rangle + |1\rangle = \sqrt{2}|+\rangle \quad (26)$$

$$\textcircled{\pi} - = |+\rangle - |-\rangle = \sqrt{2}|1\rangle \quad \textcircled{\pi} - = |0\rangle - |1\rangle = \sqrt{2}|-\rangle \quad (27)$$

We see that we get the right states, but with a wrong scalar factor. Just as we can ignore global phases in quantum circuits, we can usually ignore global non-zero scalar factors in ZX-diagrams and that is in fact what we will do throughout this paper. For a more thorough discussion on scalars in ZX-diagrams see Section 3.4.

Traditionally, Z-spiders are written as *green* dots and X-spiders as *red* dots.⁵ To accommodate colour-blindness and printing we choose not to do this, and instead to follow [31] and write Z-spiders as white dots and X-spiders as grey dots. If one wishes to use the traditional green/red colours, the author recommends using the colour-blind friendly versions listed on the ZX-calculus website.⁶ Note that the green/red or white/grey colours are not universally used: Gidney et al. [64] use black dots to represent Z-spiders and white dots to represent X-spiders (in order to match it more closely with the translation of a CNOT into the ZX-calculus).

3.2 Composition

A ZX-diagram can be built iteratively by composing other ZX-diagrams either horizontally, by connecting the output wires of the first to the input wires of the second, or vertically, simply by ‘stacking’ the diagrams to create the tensor product. The base case of this inductive construction starts with the Z- and X-spiders.

To make more clear how this works exactly, let us work through an explicit construction of the CNOT gate. The first ingredient we need is the phaseless Z-spider with 1 input and 2 outputs:

$$\text{---} \circ \begin{array}{l} \diagup \\ \diagdown \end{array} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (28)$$

Its matrix has 2 columns and 4 rows. The matrix of a general ZX-diagram with n inputs and m outputs will have 2^n columns, and 2^m rows. In particular, a ZX-diagram with no inputs is a column vector, and hence a quantum state, while a ZX-diagram with no outputs is a row vector, and hence an effect. If it has no inputs or outputs then it is a size 1×1 matrix, i.e. just a complex number.

Suppose now that we wish to horizontally compose the spider (28) with an identity, which has matrix $\text{diag}(1, 1)$. The way we calculate the result is with the Kronecker product:

$$\text{---} \circ \begin{array}{l} \diagup \\ \diagdown \end{array} \text{---} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

⁵The reason for this is that when Bob Coecke and Ross Duncan developed the ZX-calculus in 2007 they had a whiteboard with red and green markers [Duncan, personal communication].

⁶<http://zxcalculus.com/accessibility.html>

Let us now calculate the matrix of the other ingredient we need:

$$\text{Diagram: a horizontal wire with a dot and a curved line connecting it to another horizontal wire below} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Now to make a CNOT we need to horizontally compose these two subdiagrams:

$$\text{Diagram: two horizontal wires. The top wire has a dot and a curved line connecting it to the bottom wire, which has a dot. A vertical dashed line is between the two dots.} \quad (29)$$

To see that this is indeed a CNOT we calculate its matrix. On the level of the matrix, the horizontal composition of diagrams corresponds to matrix multiplication:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Up to a scalar factor of $\sqrt{2}$ this is indeed the matrix of the CNOT gate.

If this type of matrix calculation seems tedious: that is precisely the point why we want to use the ZX-calculus. We will see that many identities which would normally be proven by working with matrices can be replaced with graphical reasoning.

In the previous section we wrote the CNOT gate as

$$\text{Diagram: two horizontal wires. The top wire has a dot and a vertical line connecting it to the bottom wire, which has a dot.} \quad (30)$$

which does not match the diagram (29). In fact, in Section 3.1 we defined spiders as having inputs, which come from the left, and outputs, which come from the right. So vertical wires aren't even properly defined.

The reason we are allowed to write this wire vertically is because of the following equation (which the reader is invited to verify by showing that the matrix on the right-hand side indeed matches the matrix of the diagram on the left-hand side):

$$\text{Diagram: two horizontal wires. The top wire has a dot and a diagonal line connecting it to the bottom wire, which has a dot.} = \text{Diagram: two horizontal wires. The bottom wire has a dot and a diagonal line connecting it to the top wire, which has a dot.} \quad (31)$$

So whether the middle wire acts as an output of the Z-spider (as in the left-hand side), or as an input to the Z-spider (as in the right-hand side) does not change the matrix it represents. This means that we can write a vertical wire without ambiguity, because we can resolve it either way, and the matrix will be the same. This is an example of a broader set of symmetries that ZX-diagrams have that we will study in the next section.

For those readers familiar with tensor networks it might be helpful to note that ZX-diagrams are in fact just tensor networks. Indeed, considering them as tensor networks, our notation matches the standard notion of a graphical tensor network as introduced by

Penrose [104]. Considered as a tensor network, each wire in a ZX-diagram corresponds to a 2-dimensional index and a wire between two spiders denotes a tensor contraction. The Z- and X-spiders are then defined as:

$$(\bigcirc)_{i_1 \dots i_m}^{j_1 \dots j_n} = \begin{cases} 1 & \text{if } i_1 = \dots = i_m = j_1 = \dots = j_n = 0 \\ e^{i\alpha} & \text{if } i_1 = \dots = i_m = j_1 = \dots = j_n = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(\bigcirc)_{i_1 \dots i_m}^{j_1 \dots j_n} = \frac{1}{\sqrt{2}} \cdot \begin{cases} 1 + e^{i\alpha} & \text{if } \bigoplus_k i_k \oplus \bigoplus_l j_l = 0 \\ 1 - e^{i\alpha} & \text{if } \bigoplus_k i_k \oplus \bigoplus_l j_l = 1 \end{cases}$$

where i_k, j_l range over $\{0, 1\}$ and $\oplus$ is addition modulo 2 (i.e. XOR). While ZX-diagrams are ‘just’ tensor networks, this glosses over the most important property of ZX-diagrams which sets it apart from the tensor networks that are used in for instance condensed matter physics. Namely, that we have a set of diagrammatic rewrite rules that act on the tensor network itself.

We have been writing our diagrams with the inputs on the left and the outputs on the right in order to mimic conventional quantum circuit notation. This is however not a fixed convention for ZX-diagrams and some authors write ZX-diagrams vertically. While this is not an issue per se, what is more inconvenient is that some authors prefer time flowing upwards, while others have their time flowing downwards. Whether wires coming in from the bottom are inputs or outputs hence varies per paper. The reader is advised to check carefully for each paper which convention is used, for instance by looking at the linear map interpretation of a spider with a single wire going up (if it is a column vector then time is flowing upwards; if it is a row vector, then time is flowing downwards).

3.3 Symmetries

An important reason ZX-diagrams are a useful representation is because of the number of symmetries that are present in the generators. To state these symmetries we need to introduce a couple of more generators of ZX-diagrams.

A generator that has a natural circuit counterpart is the *swap*:

$$\begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (32)$$

The swap generators interact with spiders the way you would expect. Namely, that you can ‘slide’ a spider along the wires, such as in the equation commuting a CNOT gate through a swap below:

$$\begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} = \begin{array}{c} \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} \begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \circ \text{---} \end{array} \end{array} \quad (33)$$

We will not write down the full set of equations that are associated to the swap gate, and instead will let the reader’s intuition do the work. An important point to note though is that we are not working in a ‘braided’ setting, and hence the swap is self-inverse:

$$\begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad (34)$$

The spiders are symmetric tensors. This means that they are invariant under swapping of some of their wires:

$$\begin{aligned}
 \begin{array}{c} \text{Diagram 1: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array} &= \begin{array}{c} \text{Diagram 2: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array} = \begin{array}{c} \text{Diagram 3: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array} \\
 \begin{array}{c} \text{Diagram 4: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array} &= \begin{array}{c} \text{Diagram 5: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array} = \begin{array}{c} \text{Diagram 6: Spider with 4 wires, 2 inputs, 2 outputs, phase } \alpha \end{array}
 \end{aligned} \tag{35}$$

These symmetries hold for all phases α , for spiders with any number of input and output wires, and for the swapping of any of their inputs or outputs.

Swapping of wires is not the only symmetry ZX-diagrams possess: it also allows you to swap inputs with outputs. In order to state this symmetry we need to introduce two other generators, that we write as bended pieces of wire:

$$\left(= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right) \quad \quad \quad \left) = \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix} \tag{36}$$

The first of these, the state, we call the *cup*, while the second, the effect, we call the *cap*. The astute reader might recognise these matrices as the maximally entangled 2-qubit state, also known as the Bell state, and respectively the Bell effect. The reason we write these generators as pieces of wire is because, just as with the swap, they act like pieces of wires. In particular, they satisfy the *yanking equations*:

$$\begin{array}{c} \text{Cup diagram} \end{array} = \text{straight wire} \quad \quad \quad \begin{array}{c} \text{Cap diagram} \end{array} = \begin{array}{c} \text{Cup diagram} \end{array} \tag{37}$$

Just as we can slide spiders along the wires of a swap gate (cf. (33)), we can slide a spider along the wires of cups and caps. For instance:

$$\begin{array}{c} \text{Spider on cup wire} \end{array} = \begin{array}{c} \text{Spider on cap wire} \end{array} \quad \quad \quad \begin{array}{c} \text{Cap with dot} \end{array} = \begin{array}{c} \text{Cap with dot} \end{array} \tag{38}$$

The general symmetries for the spiders with respect to the cups and caps can be expressed as follows:

$$\begin{aligned}
 \begin{array}{c} \text{Spider on cup wire} \end{array} &= \begin{array}{c} \text{Spider on cap wire} \end{array} & \begin{array}{c} \text{Spider on cup wire} \end{array} &= \begin{array}{c} \text{Spider on cap wire} \end{array} \\
 \begin{array}{c} \text{Spider on cap wire} \end{array} &= \begin{array}{c} \text{Spider on cup wire} \end{array} & \begin{array}{c} \text{Spider on cap wire} \end{array} &= \begin{array}{c} \text{Spider on cup wire} \end{array}
 \end{aligned} \tag{39}$$

The might just look like some meaningless bending of wires *and that is exactly the point*. While the cup and cap generator represent specific matrices and must be treated as such when calculating the matrix of a ZX-diagram, when working with just the ZX-diagram we can treat cups and caps simply as pieces of wires and bend them in any way we see fit. We let the notation do the work for us.

In fact, now that we have introduced the cups and caps we can state a particularly useful property that ZX-diagrams enjoy:

Only connectivity matters

What this means is that a ZX-diagram can be deformed arbitrarily by moving the generators around in the plane, bending and unbending the wires, and as long as the order of the inputs and outputs of the diagram are preserved, it represents the same matrix. Another way to state this is that if two ZX-diagrams contain the same number of Z- and X-spiders, have the same phases, are connected by the same number of wires in each diagram, and the inputs and outputs of the diagrams are connected in the same order to the same spiders, then the ZX-diagrams represent the same matrix. Again another way to state this is that we can treat ZX-diagrams as undirected multigraphs (i.e. graphs that allow more than one edge between vertices), with some additional data on the vertices. In this representation the spiders are the vertices, and each vertex is labelled by a type (Z or X), a phase and a list of inputs and outputs it is connected to.

To make more clear the type of freedom this entails, *only connectivity matters* means that the following ZX-diagrams all represent the same matrix.

(40)

With any of these ZX-diagrams, if we wish to calculate the matrix they represent, we would first have to express it as a composition of tensor products of compositions of tensor products (and so on) of Z-spiders, X-spiders, identities, swaps, cups and caps. Alternatively, viewing the ZX-diagram as a tensor network, we can contract the tensors of the spiders according to the connectivity of the diagram using whatever tensor contraction algorithm we see fit.

3.4 Scalars

In the diagram deformation example above, the zero-arity X-spider with a $-\frac{\pi}{2}$ phase can be moved completely freely throughout the diagram. This is because a zero-arity spider, or rather, any ZX-diagram with zero inputs and outputs, represents a *scalar*, a single complex number. For instance, we have:

$$\begin{array}{ll}
 \circ & = 2 \\
 \textcircled{\pi} & = 0 \\
 \textcircled{\alpha} & = 1 + e^{i\alpha}
 \end{array}
 \qquad
 \begin{array}{ll}
 \textcircled{\alpha} \circ & = \sqrt{2} \\
 \textcircled{\alpha} \textcircled{\pi} & = \sqrt{2}e^{i\alpha} \\
 \textcircled{\alpha} \textcircled{\alpha} & = \frac{1}{\sqrt{2}}
 \end{array}
 \tag{41}$$

If we have a pair of scalar subdiagrams in a ZX-diagram then in the calculation of the matrix these combine by multiplying together the numbers they represent. Note that using just combinations of the scalar diagrams of (41) we can then represent any complex number as a scalar ZX-diagram.⁷

⁷This boils down to showing we can represent any complex number as a multiplication of $\frac{1}{\sqrt{2}}$, $\sqrt{2}e^{i\alpha}$ and $1 + e^{i\alpha}$. It suffices to show we can represent any complex number z with $|z| \leq 2$, because then for any number with larger absolute value we can first rescale it by multiplying with $\frac{1}{\sqrt{2}}$ repeatedly. We can choose an α such that $|z| = |1 + e^{i\alpha}|$. Hence $\frac{z}{1 + e^{i\alpha}} = e^{i\beta}$ for some β . Hence $z = (1 + e^{i\alpha})(\sqrt{2}e^{i\beta})\frac{1}{\sqrt{2}}$.

In this paper and in many other work dealing with the ZX-calculus scalar factors are often dropped. This is for the same reason that physicists will sometimes work with unnormalised quantum states: it is simply inconvenient and sometimes unnecessary to know the exact scalar values. Note that dropping the scalar should only be done if the scalar is non-zero, since otherwise the entire diagram should cancel in an appropriate way.

A particular case where non-zero scalar factors can be ignored is when dealing with ZX-diagrams representing unitary quantum circuits (we in fact already did this in Section 3.1 in the representation of the CNOT gate, where we suppressed a factor of $\frac{1}{\sqrt{2}}$). The reason we can ignore those factors is because we can always easily retrieve the correct factor: the matrix M is proportional to a unitary, and hence $MM^\dagger = \lambda \text{id}$ for some $\lambda \in \mathbb{R}_{>0}$. The correct scalar (up to an innocuous global phase) is then $1/\sqrt{\lambda}$. We can efficiently calculate this value λ by composing the ZX-diagram with its adjoint and simplifying it until it reduces to the identity. This is efficient, because in the adjoint the order of the gates is flipped, and hence we can cancel each gate with its inverse to reveal the constant of proportionality of the diagram representing each gate and multiplying these factors together.

A situation where knowing the exact scalar value is important is when calculating an amplitude, for instance when contracting a circuit with a particular input state and output effect.

We can often write the scalar correction just as a number next to the diagram, instead of representing it as a scalar ZX-diagram. For instance, instead of writing

$$\begin{array}{c} \bullet \text{---} \circ \\ | \\ \text{---} \circ \text{---} \\ | \\ \bullet \text{---} \end{array} \quad (42)$$

to represent a scalar-accurate CNOT gate, we can write

$$\sqrt{2} \begin{array}{c} \circ \text{---} \\ | \\ \bullet \text{---} \end{array} \quad (43)$$

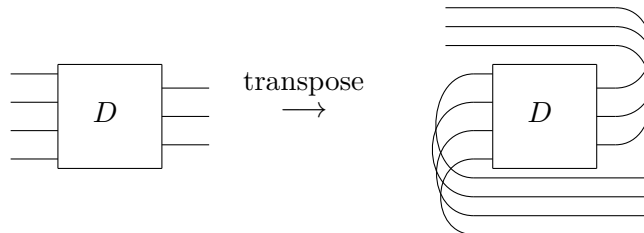
In this paper, we will ignore global scalar factors everywhere. Hence ‘=’ should be read as ‘equal up to global non-zero scalar’.

3.5 Adjoints, transpose and conjugate

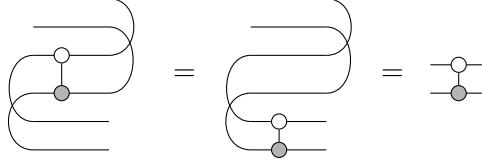
There are some useful global operations on ZX-diagrams that correspond to well-known matrix operations.

First of all, if we take a ZX-diagram and negate all the phases in the spiders, so that for instance $\frac{\pi}{2}$ is mapped to $-\frac{\pi}{2} \equiv \frac{3\pi}{2}$, and π is mapped to $-\pi \equiv \pi$, the matrix of the resulting diagram is the *conjugate* of the matrix of the original diagram (recall that the conjugate of a complex matrix is simply the pointwise complex conjugate of each the elements of the matrix).

The transpose of a matrix is also easily represented by ZX-diagrams: we use cups and caps to change every input into an output and vice versa.



Note that these ‘crossovers’ of the wires are necessary to ensure that the first input is mapped to the first output, instead of to the last output. If we apply this procedure to the CNOT gate we can verify that the CNOT gate is indeed self-transpose:



Here in the first step we slid the spiders along the wires to the bottom, and in the second step we applied the first yanking equation (37).

The adjoint of a matrix is constructed by taking the transpose and conjugating the matrix. Hence, the adjoint of a ZX-diagram is constructed by interchanging the inputs and outputs using cups and caps as described above, and then negating all the phases of the spiders in the diagram.

3.6 Hadamards

In Section 2 we used some special notation for the Hadamard gate:

$$\text{---}\square\text{---} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

We can either decide to add this ‘Hadamard box’ as a new generator to the ZX-calculus or by seeing it as a *derived* generator defined in terms of the other generators. We will take the latter route.⁸

Recall that any single-qubit unitary gate is equal (up to global phase) to a rotation of the Bloch sphere over some axis. Hence, by decomposing the rotation using Euler angles we can represent any single-qubit unitary with a Z-rotation followed by an X-rotation, followed once again by a Z-rotation:

$$\text{---}\square\text{---} = \text{---}\alpha\text{---}\beta\text{---}\gamma\text{---}$$

In the particular case of $U = H$, we can take $\alpha = \beta = \gamma = \frac{\pi}{2}$:

$$\text{---}\square\text{---} = e^{-i\frac{\pi}{4}} \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} \quad (44)$$

The global phase of $e^{-i\frac{\pi}{4}}$ is included in order to make the equation exactly equal. If one is willing to accept these global scalar factors as an inherent part of the ZX-calculus this definition is fine. On a formal level it is however desirable to have a representation that does not need scalar corrections:

$$\text{---}\square\text{---} = \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} \quad (45)$$

Note that these are by no means the only ways to represent the Hadamard gate using spiders. There is in fact an entire family of representations that will also be useful to note:

$$\begin{aligned} \text{---}\square\text{---} &= e^{-i\frac{\pi}{4}} \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} = e^{i\frac{\pi}{4}} \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} = \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} \\ &= e^{-i\frac{\pi}{4}} \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} = e^{i\frac{\pi}{4}} \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} = \text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---}\frac{\pi}{2}\text{---} \end{aligned}$$

⁸For practical purposes it makes no difference whether the Hadamard is a generator or is simply syntactic sugar. However, when studying completeness of the ZX-calculus (cf. Section 9) or when defining non-standard interpretations of ZX-diagrams, it is relevant how the Hadamard is defined.

Using any of these definitions it is easy to show that the Hadamard box in the ZX-calculus is self-transpose:

$$\begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{\text{H}} \begin{array}{c} \text{---} \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \boxed{\text{H}} \begin{array}{c} \text{---} \\ \text{---} \end{array} \quad (46)$$

This means that, just like with the spiders, the orientation of the Hadamard is not important, which is why we are allowed to draw it as a simple box. Also just like with the spiders, we can slide Hadamards along wires as intuition would suggest.

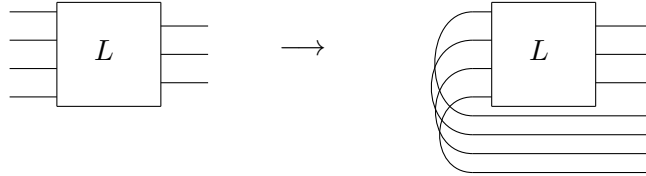
3.7 Universality

We have now seen that quite a number of quantum gates and constructions can be represented in the ZX-calculus. This raises the following question: exactly which linear maps can be represented in the ZX-calculus. The answer to this is simple: *everything*. Here, ‘everything’ means any complex matrix of size $2^n \times 2^m$ for some n and m .

We have already seen how to represent an arbitrary complex number using a scalar ZX-diagram (Section 3.4). We have also seen that we can represent arbitrary Z and X rotations of the Bloch sphere (Eqs. (23) and (24)). These rotations generate all single-qubit unitaries (up to global phase, which we can reconstruct by adding some scalar diagram). We have also seen that we can construct CNOT gates (see (29)). The single-qubit unitaries together with the CNOT gate form a universal gate set, and hence any unitary on an arbitrary number of qubits can be written as a composition of these gates. So we see that we can indeed represent any unitary between qubits as a ZX-diagram.

For any n -qubit normalised quantum state $|\psi\rangle$ there exists a n -qubit unitary U such that $U|0\cdots 0\rangle = |\psi\rangle$. As we can represent both $|0\cdots 0\rangle$ and U as a ZX-diagram, this means that we can represent an arbitrary normalised quantum state as a ZX-diagram. By composing it with the appropriate scalar diagram we can rescale this to a vector of arbitrary norm.

Now given some linear map L from n qubits to m qubits we can transform it into a $(n + m)$ -qubit state by the Choi-Jamiołkowski isomorphism:



This state can be represented as a ZX-diagram, and hence by bending back the wires using the yanking equation (37) we get a representation of L as a ZX-diagram.

The property of ZX-diagrams that they can represent any linear map is known as the *universality* of the ZX-calculus. Together with two other properties we will encounter later, soundness and completeness, they form the backbone of the usefulness of ZX-diagrams.

Note that just because we *can* represent any linear map using a ZX-diagram, this does not mean that such representations will necessary be ‘nice’. It is an ongoing challenge to find good representations of useful linear maps in the ZX-calculus. For instance, in Section 8 we will introduce a new (derived) generator that allows us to more easily represent Toffoli gates in the ZX-calculus, since the native representation is rather cumbersome.

4 The ZX-calculus

In the previous section we introduced ZX-diagrams as essentially a graphical notation for complex matrices of size $2^n \times 2^m$. In this section we will see how we can actually

do calculations with ZX-diagrams, transforming it from notation into a language. Each subsection here introduces new rewrite rules and demonstrates how these rewrite rules can be used to prove well-known properties of quantum circuits and states. We summarise all the rules in Figure 1 in Section 4.6.

4.1 Spider fusion and identity removal

The most fundamental of all the graphical rewrite rules allowed in the ZX-calculus is *spider fusion*. This says that when two spiders of the same colour are connected by one or more wires, that we can *fuse* these two spiders together. When spiders are fused, their phases are added together. In terms of diagrams:

(47)

Note that we interpret the numbers α and β as phases $e^{i\alpha}$ and $e^{i\beta}$, and hence this addition is assumed to be modulo 2π .

We already saw a few instances of spider fusion in Section 2, such as when proving that a T gate commutes through the control of a CNOT:

Flipping the colours, we can use spider fusion to show that a X gate commutes through the target of a CNOT gate:

The adding of the phases essentially generalises the fact that two rotations of the Bloch sphere in the same direction add together:

$$\text{---} \bigcirc_{\alpha} \text{---} \bigcirc_{\beta} \text{---} = \text{---} \bigcirc_{\alpha+\beta} \text{---} \quad (48)$$

Recalling that in the ZX-calculus $|0\rangle = \textcircled{\bullet}$ and $|1\rangle = \textcircled{\pi}$ (cf. (26)), we also see that spider fusion gives us that applying the Pauli X gate to $|0\rangle$ gives $|1\rangle$:

$$\text{small circle} - \text{large circle}(\pi) - \text{line} = \text{large circle}(\pi) - \text{line} \quad (49)$$

An equation like $Z|-\rangle = |+\rangle$ is given similarly:

$$\text{---}(\pi)\text{---}(\pi)\text{---} = \text{---}(\bigcirc)\text{---} \quad (50)$$

Note here that we have $\pi + \pi = 2\pi \equiv 0$.

In fact, the spider fusion equations (47) states something much stronger than is evident from these particular cases: any family of linear maps satisfying an analogous set of equations is completely characterised by an orthonormal basis [38], cf. Section 7.5.

Another rewrite rule we already saw is *identity removal*:

$$\text{---}\bigcirc\text{---} = \text{---} = \text{---}\bullet\text{---} \quad (51)$$

This can be interpreted in several ways. Considering a 1-input 1-output spider as a phase gate over either the Z- or X-axis, this rule says that a rotation by 0 degrees does nothing. Combining it with spider fusion it says that the inverse of a rotation by α is a rotation by $-\alpha$:

$$\text{---}(\alpha)\text{---}(\alpha)\text{---} = \text{---}\bigcirc\text{---} = \text{---}$$

Composing the diagrams of (51) with a cup it says we can make a Bell state by either taking a maximally entangled state over the Z basis or over the X basis:

$$|00\rangle + |11\rangle = \text{---}\bigcirc\text{---} = \text{---}\big(\text{---}\big) = \text{---}\big(\bullet\text{---}\big) = |++\rangle + |--\rangle$$

Identity removal also allows us to get rid of self-loops on spiders:

$$\begin{array}{c} \vdots \\ \text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} \stackrel{(51)}{=} \begin{array}{c} \text{---}\big(\bullet\big)\text{---} \\ \vdots \\ \text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} \stackrel{(47)}{=} \begin{array}{c} \vdots \\ \text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} \quad (52)$$

Here in the first step we applied the rule in reverse to *add* an identity. Then we fused the two spiders that are now connected by two wires.

4.2 The copy rule and π -commutation

The second set of rules of the ZX-calculus we will look at concerns the interaction of the Pauli X and Z gates and their eigenstates $|0\rangle, |1\rangle, |+\rangle, |-\rangle$ with the spiders. Recall from Eq. (26) that (up to global non-zero scalar factors):

$$\begin{array}{ll} \bullet\text{---} = |0\rangle & \bigcirc\text{---} = |+\rangle \\ \pi\text{---} = |1\rangle & \pi\text{---} = |-\rangle \\ \text{---}\pi\text{---} = X & \text{---}\pi\text{---} = Z \end{array}$$

The spider fusion rule already covers the interaction of a couple of these maps and states. Indeed we can use the spider fusion rule to show that a Z gate commutes through a Z-spider:

$$\begin{array}{c} \pi \\ \vdots \\ \text{---}\big(\pi\big)\text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} = \begin{array}{c} \text{---}\big(\alpha + \pi\big)\text{---} \\ \vdots \end{array} = \begin{array}{c} \text{---}\big(\alpha\big)\text{---}\big(\pi\big)\text{---} \\ \vdots \end{array}$$

Spider fusion also shows what happens when a $|+\rangle$ or $|-\rangle$ state is applied to a Z-spider:

$$\begin{array}{c} \bigcirc \\ \vdots \\ \text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} = \begin{array}{c} \text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} \quad \begin{array}{c} \pi \\ \vdots \\ \text{---}\big(\pi\big)\text{---}\big(\alpha\big)\text{---} \\ \vdots \end{array} = \begin{array}{c} \text{---}\big(\alpha + \pi\big)\text{---} \\ \vdots \end{array}$$

By flipping all the colours we then also know that an X gate commutes through an X spider and that the $|0\rangle$ and $|1\rangle$ fuse into an X spider.

A more interesting question is what happens to an X gate when it encounters a Z-spider (or vice versa, a Z gate that encounters an X-spider). Consider a Z-spider with just a single input and no phase. Its linear operator is then given by $|0 \cdots 0\rangle\langle 0| + |1 \cdots 1\rangle\langle 1|$. Hence, if we apply an X gate to the input the resulting linear operator is

$$|0 \cdots 0\rangle\langle 0| X + |1 \cdots 1\rangle\langle 1| X = |0 \cdots 0\rangle\langle 1| + |1 \cdots 1\rangle\langle 0|,$$

as an X gate interchanges $|0\rangle$ and $|1\rangle$. This is easily seen to be equivalent to a Z-spider *followed* by an X gate on each of the outputs:

$$|0 \cdots 0\rangle\langle 1| + |1 \cdots 1\rangle\langle 0| = (X \otimes \cdots \otimes X) |1 \cdots 1\rangle\langle 1| + (X \otimes \cdots \otimes X) |0 \cdots 0\rangle\langle 0|.$$

Hence, in terms of diagrams, we have:

$$\text{---} \pi \text{---} \bigcirc \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \text{---} \bigcirc \begin{array}{c} \pi \\ \pi \\ \vdots \end{array} \quad (53)$$

We will refer to this as a π -copy rule. By applying the appropriate cups and caps it should be clear that such a π -copy rule continues to hold regardless of how many inputs and outputs the spider has. But what about when the spider has a non-zero phase? Using spider fusion in reverse to *unfuse* the phase we can reduce this to a simpler question:

$$\text{---} \pi \text{---} \alpha \text{---} \bigcirc \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} \stackrel{(47)}{=} \text{---} \bigcirc \begin{array}{c} \alpha \\ \pi \\ \vdots \end{array} \stackrel{(53)}{=} \text{---} \bigcirc \begin{array}{c} \alpha \pi \\ \pi \\ \vdots \end{array} \quad (54)$$

Hence, we need to resolve what happens when we apply an X gate to a $|0\rangle + e^{i\alpha} |1\rangle$ state:

$$X(|0\rangle + e^{i\alpha} |1\rangle) = X|0\rangle + e^{i\alpha} X|1\rangle = |1\rangle + e^{i\alpha} |0\rangle = e^{i\alpha}(|0\rangle + e^{-i\alpha} |1\rangle).$$

Ignoring the global phase $e^{i\alpha}$ we can write this in terms of a diagram:

$$\alpha \text{---} \pi \text{---} = \text{---} \alpha \text{---} \quad (55)$$

The most generic case, together with its colour-flipped counterpart is then:

$$\begin{array}{c} \pi \\ \vdots \end{array} \text{---} \alpha \text{---} \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \begin{array}{c} \pi \\ \vdots \end{array} \text{---} \alpha \text{---} \begin{array}{c} \pi \\ \vdots \end{array} \quad \begin{array}{c} \pi \\ \vdots \end{array} \text{---} \alpha \text{---} \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \begin{array}{c} \pi \\ \vdots \end{array} \text{---} \alpha \text{---} \begin{array}{c} \pi \\ \vdots \end{array} \quad (56)$$

Similar copy rules hold for the eigenstates of the X and Z operators. Indeed, applying a $|0\rangle$ to the 1-input Z-spider $|0 \cdots 0\rangle \langle 0| + e^{i\alpha} |1 \cdots 1\rangle \langle 1|$ it is clear that we get the output $|0 \cdots 0\rangle$:

$$\bullet \text{---} \alpha \text{---} \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} \quad (57)$$

The same works for $|1\rangle$ (although we then do get a global phase of $e^{i\alpha}$ that we will ignore):

$$\pi \text{---} \alpha \text{---} \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \begin{array}{c} \pi \\ \pi \\ \vdots \end{array} \quad (58)$$

By introducing a Boolean variable a that is either 0 or 1 we can represent the last two equations as a single parametrised equation:

$$a\pi \text{---} \alpha \text{---} \begin{array}{c} \text{---} \\ \text{---} \\ \vdots \end{array} = \begin{array}{c} a\pi \\ a\pi \\ \vdots \end{array} \quad (59)$$

Here the $a\pi$ phase is either zero (when $a = 0$), or π (when $a = 1$), and hence the input represents either a $|0\rangle$ or a $|1\rangle$.

As before, the same equation but with the colours flipped also continues to hold, which gives rules for copying $|+\rangle$ and $|-\rangle$ states through X-spiders. We will refer to all these rules as *state-copy rules*. These state-copy rules only hold when the spider being copied has a phase of 0 or π . For any other phase the analogue of (59) does not hold.

Note that the orientation of the wires in the rewrite rules we have derived is irrelevant, as we can pre-compose and post-compose each rule with ‘cups and caps’ to change inputs to outputs and vice versa. For instance, for the ‘reverse’ of the state-copy rule:

$$\text{Diagram (60)} \quad (60)$$

At this point the reader might wonder why we focus so much on X and Z, and not on Y. The reason for this is that Y can be presented as a composition of X and Z, and hence the rules relating to the Y-eigenbasis can be derived from the equations we have covered here. In principle, we could have worked with the ‘XY’-calculus instead of the ZX-calculus, but there is an important reason to prefer Z and X over Y. Namely, the Z and X eigenbases are *self-conjugate* meaning that each of the vectors in the basis is equal to its own componentwise complex conjugate, for instance: $\overline{|0\rangle} = |0\rangle$. This is not the case for the Y eigenbasis $|\pm i\rangle := |0\rangle \pm i|1\rangle$. As a result, the ‘Y-spider’ does not enjoy the symmetry between inputs and outputs of (39) that the Z- and X-spider do.

4.3 Formal rewriting and soundness

At this point it might be helpful to say a bit more about the formal nature of the rewriting we are doing. The rewrite rules we have given above are of the form $D_1 = D_2$, where D_1 and D_2 are some ZX-diagrams with the same number of inputs and outputs. We then used these rewrite rules on *larger* diagrams that contain either D_1 or D_2 as subdiagrams. This is because when we are asserting $D_1 = D_2$, we are actually asserting an entire family of equalities. We are for instance also asserting that $D_1 \otimes E = D_2 \otimes E$ for any other ZX-diagram E , and $D \circ D_1 = D \circ D_2$ for any composable ZX-diagram D .

In a sense we have the following implication:

$$\text{Diagram (61)} \quad (61)$$

I.e. if D_1 appears as a subdiagram of some larger ‘ambient’ diagram D' , then D_1 can be replaced by D_2 inside of D' .

An important question to settle here is whether this is *sound*. A rewrite rule in a language is called sound when it preserves the underlying semantics. In the setting of ZX-diagrams, the language is the set of diagrammatic rewrite rules of the ZX-calculus, and the semantics of a ZX-diagram is the linear map it represents. So a rule in the ZX-calculus is sound when the diagrams on each side of the equation represent the same matrix. For a single equation like $D_1 = D_2$ soundness is easy to check: just calculate the matrix on both sides. But how can we be sure that employing this rewrite rule in a larger context as in (61) is still sound, as this concerns an infinite family of rewrite rules?

To make the following discussion more clear, we will adopt the convention of the literature on the ZX-calculus to denote the matrix of a ZX-diagram D by $\llbracket D \rrbracket$. The soundness of a rule $D_1 = D_2$ is then witnessed by the equation $\llbracket D_1 \rrbracket = \llbracket D_2 \rrbracket$.

By definition of how we calculate the matrix of a ZX-diagram, we have $\llbracket D_1 \otimes D_2 \rrbracket = \llbracket D_1 \rrbracket \otimes \llbracket D_2 \rrbracket$ and $\llbracket D_1 \circ D_2 \rrbracket = \llbracket D_1 \rrbracket \llbracket D_2 \rrbracket$ (where this latter juxtaposition represents matrix multiplication). Hence, if we have $\llbracket D_1 \rrbracket = \llbracket D_2 \rrbracket$, then we also have $\llbracket D_1 \otimes E \rrbracket = \llbracket D_1 \rrbracket \otimes \llbracket E \rrbracket = \llbracket D_2 \rrbracket \otimes \llbracket E \rrbracket = \llbracket D_2 \otimes E \rrbracket$, and hence $D_1 \otimes E = D_2 \otimes E$ is also sound. The analogous result for composition with E also holds. As ZX-diagrams are built iteratively by repeated tensor products and composition, this shows that (61) is indeed sound.

In terms of category theory, we can frame $\llbracket \cdot \rrbracket$ as a strong monoidal functor from the category of ZX-diagrams to the category of complex matrices of size 2^n , see Section 7 for more details. The type of local rewriting we do can be understood as an instance of *double-pushout rewriting* [18].

4.4 Colour changing

As has been noted several times in the preceding sections, when we have derived some rule in the ZX-calculus, an analogous rule with the colours interchanged also holds. This follows from the rules concerning the Hadamard gate that we introduced in Section 2.3. Let us repeat these rules here for ease of reference:

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \\ \vdots \\ \text{---} \square \end{array} = \text{---} \quad \quad \quad \begin{array}{c} \text{---} \square \\ \vdots \\ \text{---} \square \end{array} \alpha \begin{array}{c} \text{---} \square \\ \vdots \\ \text{---} \square \end{array} \stackrel{(14)}{=} \begin{array}{c} \text{---} \square \\ \vdots \\ \text{---} \square \end{array} \alpha \begin{array}{c} \text{---} \square \square \\ \vdots \\ \text{---} \square \square \end{array} \stackrel{(16)}{=} \begin{array}{c} \text{---} \square \\ \vdots \\ \text{---} \square \end{array} \alpha \begin{array}{c} \text{---} \square \\ \vdots \\ \text{---} \square \end{array} \quad (62)$$

The first of these rules states that the Hadamard gate is self-inverse, while the second says that commuting a Hadamard through a spider changes its colour (reflecting the fact that the Hadamard interchanges the eigenbasis of the Z and X gate). Note that the Hadamard commutation rule also holds for spiders that have no inputs, so for instance:

$$\alpha \text{---} = \alpha \text{---} \square \text{---} \quad (63)$$

Let us give an example to demonstrate how the two Hadamard rules (62) imply the colour-inverse of any other rule we have. For instance, suppose we want to prove the colour-inverse of Eq. (53):

$$\begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} = \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \quad (64)$$

We start with the left-hand side of the colour-swapped version, and then insert two Hadamards on the input by applying the self-inverse rule in reverse:

$$\begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \stackrel{(62)}{=} \begin{array}{c} \text{---} \square \square \\ \vdots \\ \text{---} \square \square \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array}$$

We then commute one of the Hadamards all the way to the outputs:

$$\begin{array}{c} \text{---} \square \square \\ \vdots \\ \text{---} \square \square \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \stackrel{(62)}{=} \begin{array}{c} \text{---} \square \square \\ \vdots \\ \text{---} \square \square \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array} \stackrel{(62)}{=} \begin{array}{c} \text{---} \square \square \\ \vdots \\ \text{---} \square \square \end{array} \alpha \begin{array}{c} \text{---} \pi \\ \vdots \\ \text{---} \pi \end{array}$$

We now see the left-hand side of the original (not colour-swapped) equation (64), and hence we can apply it:

$$\text{Diagram (64)} = \text{Diagram (64)}$$

Now it remains to commute the Hadamards back to the left, and to cancel the resulting double Hadamard:

$$\text{Diagram (62)} = \text{Diagram (62)} = \text{Diagram (62)} = \text{Diagram (62)}$$

We have hence succeeded in proving the colour-inverse of Eq. (64):

$$\text{Diagram} = \text{Diagram}$$

4.5 The bialgebra and Hopf rule

The previous sets of rules all have a very distinct topologically intuitive character: spider fusion allows you to fuse adjacent spiders of the same colour; identity removal and the Hadamard self-inverse rule allow you to remove certain vertices; the π -copy, state-copy and colour-change rule allow you to commute certain generators through spiders by copying them to the other side. It is therefore relatively easy (after you gain some practice) to spot where they can be applied and what the effect of their application will be on the structure of the rest of the diagram.

The last major rule of the ZX-calculus takes more time to work with intuitively, although it does correspond to a natural and crucial property of the interaction of the Z- and X-spider.

Before we introduce the rule, let us give some motivation. Treating the $|0\rangle$ and $|1\rangle$ states as Boolean bits, we can view the phaseless 1-input, n -output Z-spider as the COPY gate that copies the bit 0 to $0 \cdots 0$ and the bit 1 to $1 \cdots 1$. Indeed, this is exactly what Eq. (59) states. Analogously, we can view the phaseless 2-input, 1-output X-spider as the XOR gate:

$$\text{Diagram} = \text{Diagram}$$

The XOR gate and the COPY gate have a natural commutation relation. Indeed, first XORing bits, and then copying the outcome is the same as first copying each of the bits and then XORing the resulting pairs:

$$\text{Diagram (65)} = \text{Diagram (65)} \quad (65)$$

This equation says that the XOR *algebra* and the COPY *coalgebra* together form a *bialgebra* (for more about these algebraic structures see Section 7.5). This is why we refer

to the analogous equation in the ZX-calculus as the *bialgebra rule*:

$$\begin{array}{c} \text{---} \bullet \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \circ \text{---} \end{array} = \begin{array}{c} \text{---} \circ \text{---} \bullet \text{---} \\ \text{---} \circ \text{---} \bullet \text{---} \end{array} \quad (66)$$

This rule is essential to many proofs in the ZX-calculus. As a demonstration of its utility, let us prove the well-known ‘three CNOTs make a swap’ circuit identity. This proof will also reveal the need for one additional rule that can actually be derived with the bialgebra rule. We start with the three CNOTs and deform the diagram to demonstrate where the bialgebra rule can be applied:

$$\begin{array}{c} \text{---} \circ \text{---} \bullet \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \circ \text{---} \bullet \text{---} \end{array} = \begin{array}{c} \text{---} \circ \text{---} \bullet \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \circ \text{---} \bullet \text{---} \end{array} \quad (67)$$

Now, apply Eq. (66) from right-to-left, and then deform the diagram again to bring it to a cleaner form:

$$\begin{array}{c} \text{---} \circ \text{---} \bullet \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \circ \text{---} \bullet \text{---} \end{array} \stackrel{(66)}{=} \begin{array}{c} \text{---} \bullet \text{---} \circ \text{---} \bullet \text{---} \\ \text{---} \circ \text{---} \bullet \text{---} \circ \text{---} \end{array} = \begin{array}{c} \text{---} \circ \text{---} \bullet \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \circ \text{---} \bullet \text{---} \end{array} \quad (68)$$

To finish the proof it now remains to show that two CNOTs applied in succession cancel each other:

$$\begin{array}{c} \text{---} \circ \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \bullet \text{---} \end{array} = \text{---} \quad (69)$$

The first step towards doing this might seem clear, namely, we fuse the adjacent spiders of the same colour:

$$\begin{array}{c} \text{---} \circ \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \bullet \text{---} \end{array} \stackrel{(47)}{=} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \bullet \text{---} \end{array} \quad (70)$$

But now we are seemingly stuck.

The solution comes from another equation between the Boolean gates XOR and COPY. When we first apply an XOR and then a COPY, we can commute the XOR by essentially ‘copying’ it through the COPY gate. But what happens when we first apply a COPY *and then* an XOR gate to the two outputs of the COPY: the input 0 first gets copied to 00 and then XORed to 0, while the input 1 first gets copied to 11 and then XORed to 0. So regardless of the input, we output a 0. Generalising the notion of a XOR gate to allow for a 0-input XOR gate that just outputs its unit (namely 0), and a 0-output COPY gate that discards its output we can write this relation diagrammatically as:

$$\boxed{\text{COPY}} \text{---} \boxed{\text{XOR}} \text{---} = \text{---} \boxed{\text{COPY}} \boxed{\text{XOR}} \text{---} \quad (71)$$

An algebra (XOR) and a coalgebra (COPY) satisfying this equation are known together as a *Hopf algebra*. This is why we refer to the following analogous equation between the Z- and X-spider as the *Hopf rule*:

$$\text{---} \circ \text{---} \bullet \text{---} = \text{---} \circ \text{---} \bullet \text{---} \quad (72)$$

Operationally, this equation can be interpreted as saying that the Z-basis and X-basis correspond to *complementary* observables, where having maximal information about the

Z observable gives minimal information about the X observable [31, Section 9.2]. In this same vein the bialgebra rule (66) says that the observables are *strongly complementary* [37].

It turns out that complementarity can be derived from strong complementarity using some clever deformation of the diagram and the rules from the previous sections [52]:

$$\begin{array}{ccccccc}
 \text{---} \circ \text{---} & = & \text{---} \circ \text{---} & \stackrel{(51)}{=} & \text{---} \circ \text{---} & \stackrel{(47)}{=} & \text{---} \circ \text{---} \\
 & & & & & & \\
 & \stackrel{(66)}{=} & \text{---} \circ \text{---} & \stackrel{(59)}{=} & \text{---} \circ \text{---} & = & \text{---} \circ \text{---}
 \end{array}$$

In this derivation, we first deformed the diagram, then we introduced identities, we unfused some spiders, applied the bialgebra rule, copied a state twice, and in the last equation we ignored the dangling scalar diagram as discussed in Section 3.4.

The Hopf rule is exceedingly useful. Indeed, combining it with some spider fusion we can use it to show that we can always cancel pairs of connections between spiders of opposite colours:

$$\begin{array}{ccccccc}
 \begin{array}{c} \vdots \\ \alpha \\ \vdots \end{array} & \stackrel{(47)}{=} & \begin{array}{c} \vdots \\ \alpha \\ \vdots \end{array} & \stackrel{(72)}{=} & \begin{array}{c} \vdots \\ \alpha \\ \vdots \end{array} & \stackrel{(47)}{=} & \begin{array}{c} \vdots \\ \alpha \\ \vdots \end{array} \\
 \begin{array}{c} \vdots \\ \beta \\ \vdots \end{array} & & \begin{array}{c} \vdots \\ \beta \\ \vdots \end{array} & & \begin{array}{c} \vdots \\ \beta \\ \vdots \end{array} & & \begin{array}{c} \vdots \\ \beta \\ \vdots \end{array}
 \end{array} \quad (73)$$

So if two spiders of opposite colour are connected by n wires, then this can be reduced to $n \bmod 2$ wires. In particular, we can finish the proof that 2 CNOTs cancel each other out, and thus that 3 CNOTs make a SWAP:

$$\begin{array}{ccccccc}
 \begin{array}{c} \circ \\ \vdots \end{array} & \stackrel{(47)}{=} & \begin{array}{c} \circ \\ \vdots \end{array} & \stackrel{(73)}{=} & \begin{array}{c} \circ \\ \vdots \end{array} & \stackrel{(51)}{=} & \begin{array}{c} \circ \\ \vdots \end{array} \\
 \begin{array}{c} \bullet \\ \vdots \end{array} & & \begin{array}{c} \bullet \\ \vdots \end{array} & & \begin{array}{c} \bullet \\ \vdots \end{array} & & \begin{array}{c} \bullet \\ \vdots \end{array}
 \end{array} \quad (74)$$

We can extend the bialgebra rule (66) to spiders of arbitrary arity:

$$n \left\{ \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} \right\} m = n \left\{ \begin{array}{c} \vdots \\ \circ \\ \vdots \end{array} \right\} m \quad (75)$$

That is, for any two connected phaseless spiders of different colours we can apply the bialgebra rule, resulting in a fully connected bipartite graph as on the right-hand side. For $n = m = 2$ this is exactly Eq. (66). For $n = 1$ or $m = 1$ this follows in a trivial manner by adding and removing identities (cf. Eq. (51)). When $n = 0$ or $m = 0$ this is just the state-copy rule, i.e. (the colour-swapped) Eq. (57). Now suppose $n = 2$ and $m = 3$, we will show how to derive its bialgebra rule using just spider fusion and the $n = 2, m = 2$ bialgebra rule:

$$\begin{array}{ccccccc}
 \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} & \stackrel{(47)}{=} & \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} & \stackrel{(66)}{=} & \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} & & \\
 & & & & & & \\
 & \stackrel{(66)}{=} & \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} & \stackrel{(47)}{=} & \begin{array}{c} \vdots \\ \bullet \\ \vdots \end{array} & &
 \end{array} \quad (76)$$

We leave it as an exercise for the reader how to derive the cases for larger n and m (hint: use induction or see [31, Theorem 9.71]).

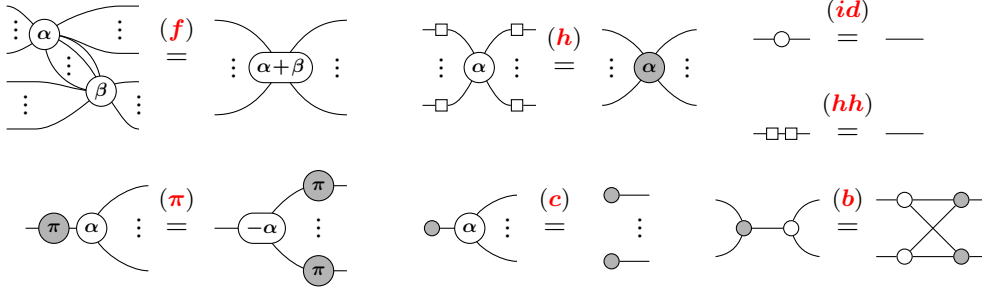


Figure 1: A convenient presentation of the ZX-calculus. These rules hold for all $\alpha, \beta \in \mathbb{R}$, and due to **(h)** and **(hh)** all rules also hold with the colours interchanged. Note ‘...’ should be read as ‘0 or more’, hence the spiders on the left-hand side of **(f)** are connected by one or more wires. The letters stand respectively for spider-**(f)**usion, **(h)**adamard, **(id)**entity, **(hh)**-cancellation, **(π)**-commute, **(c)**opy, and **(b)**ialgebra. Note that these rules are only correct up to non-zero scalar.

There are two common mistakes people make when using the bialgebra rule. The first is that the bialgebra rule (66) only works when the phases on the spiders are zero. When the phases are π a modification is possible by combining it with the π -copy rules (56):

$$\text{Diagram with two spiders } a\pi \text{ and } b\pi = \text{Diagram with two spiders } b\pi \text{ and } a\pi \quad (77)$$

But when one of the spiders contains some arbitrary phase α , the result will be more complicated:

$$\text{Diagram with spider } \alpha \text{ and a wire} \stackrel{(47)}{=} \text{Diagram with spider } \alpha \text{ and a wire with a phase pi circle} \stackrel{(75)}{=} \text{Diagram with spider } \alpha \text{ and a wire with a phase pi circle and a phase pi circle} \stackrel{(75)}{=} \text{Diagram with spider } \alpha \text{ and a wire with a phase pi circle and a phase pi circle} \quad (78)$$

The additional bit of diagram we get is called a *phase gadget* and is studied in more detail in Section 5.6.

The second common mistake is that people apply the rule (66) from right-to-left without paying attention to how many outputs the spiders have, wrongfully equating the following diagrams:

$$\text{Diagram with two spiders and a wire} \neq \text{Diagram with two spiders and a wire} \quad (79)$$

The correct way to apply the bialgebra rule here is to first unfuse the spider:

$$\text{Diagram with two spiders and a wire} \stackrel{(47)}{=} \text{Diagram with two spiders and a wire} \stackrel{(66)}{=} \text{Diagram with two spiders and a wire} \quad (80)$$

4.6 The complete calculus

We now have all the main ingredients for reasoning with ZX-diagrams using the ZX-calculus, so let us summarise what we have already seen. The rules of the ZX-calculus are presented in Figure 1.

The Hadamard is defined as any one of the following diagrams (that can be shown to

be equivalent using the rules of Figure 1):

$$\begin{aligned}
 \text{---}\square\text{---} &= e^{-i\frac{\pi}{4}} \text{---}\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\text{---} = e^{i\frac{\pi}{4}} \text{---}\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\text{---} = \text{---}\left(\frac{\pi}{2}\right)\text{---}\left(\frac{\pi}{2}\right)\text{---}\left(\frac{\pi}{2}\right)\text{---} \\
 &= e^{-i\frac{\pi}{4}} \text{---}\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\text{---} = e^{i\frac{\pi}{4}} \text{---}\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\left(\frac{\pi}{2}\right)\text{---} = \text{---}\left(\frac{\pi}{2}\right)\text{---}\left(\frac{\pi}{2}\right)\text{---}\left(\frac{\pi}{2}\right)\text{---}
 \end{aligned}$$

Besides these concrete rules we have several ‘meta-rules’:

- Only connectivity matters: diagrams can be deformed in any way, as long as the order of the inputs and outputs of the whole diagram is preserved.
- Each of the rules in Figure 1 also holds with the inputs and outputs interchanged.
- Every rule also holds with the colours (white and grey) interchanged.
- Every rule holds with all the phases negated.

We have also seen several combined and derived rules:

- The rules (c) and (b) can be combined and generalised to the following rule:

$$n \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} \text{---} \left(\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right) = n \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} \text{---} \left(\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right)$$

- The rules of Figure 1 imply the Hopf rule:

$$\text{---}\bigcirc\text{---}\bigcirc\text{---} = \text{---}\bigcirc\text{---}\bigcirc\text{---}$$

- The rules (π) and (c) can be combined to give a more generic state-copy rule:

$$\begin{array}{c} \text{---}\bigcirc\text{---} \\ \text{---}\bigcirc\text{---} \\ \vdots \\ \text{---}\bigcirc\text{---} \end{array} = \begin{array}{c} \text{---}\bigcirc\text{---} \\ \text{---}\bigcirc\text{---} \\ \vdots \\ \text{---}\bigcirc\text{---} \end{array}$$

Note that in the remainder of the paper we will often label steps in a derivation with the rule names of Figure 1 to denote which are applied. See for example (81).

The rules we present in Figure 1 are not a minimal set. A smaller set of equivalent rules can be found in for instance [31, Definition 9.108] and a ruleset that contains the correct scalars can be found in [12].

5 Some example derivations

Let us now use the ZX-calculus to reason about several small examples that demonstrate its use in reasoning about quantum computing. This will also demonstrate several best practices when trying to simplify a diagram using the ZX-calculus. A reader wishing to try out some diagrammatic proving themselves could take a look at Appendix A.6 and try to prove these circuit identities by hand (although it might be useful to read Section 5.6 first).